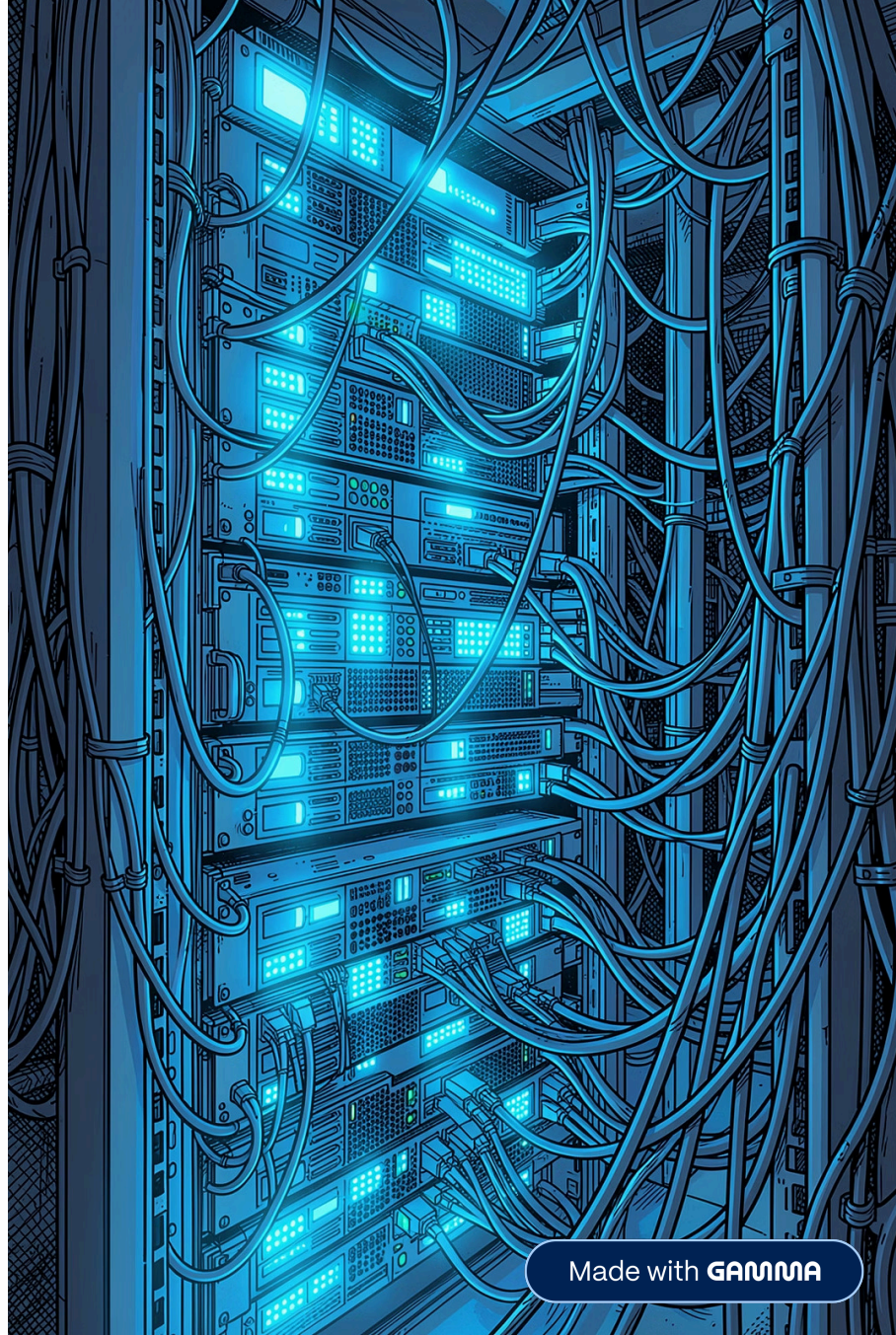
BACKEND ENGINEERING

INFOSEC STUDENTS

Containerization & Docker Fundamentals

A practical guide to reproducible environments, modern deployment workflows, and backend engineering practices — built for developers who need things to work the same way everywhere.



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The Problem Before Docker

Classic Scenario

Developer A: ✓ **Works on my machine.**
Developer B: ✗ **Cannot run it.**

Common Causes

- Different OS versions
- Mismatched language runtimes (e.g., Go 1.20 vs 1.24)
- Missing or conflicting library versions
- Undocumented local configuration

How can we make software run consistently everywhere?

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What Is a Container?

A container packages everything your application needs to run — eliminating environment mismatch.

Application Code

Your compiled binary or source

Runtime

Go, Node, Python — exact version

Dependencies

Libraries and system packages

Configuration

Environment variables, ports

Build once. Run anywhere. Same behavior on your laptop, CI, and production.

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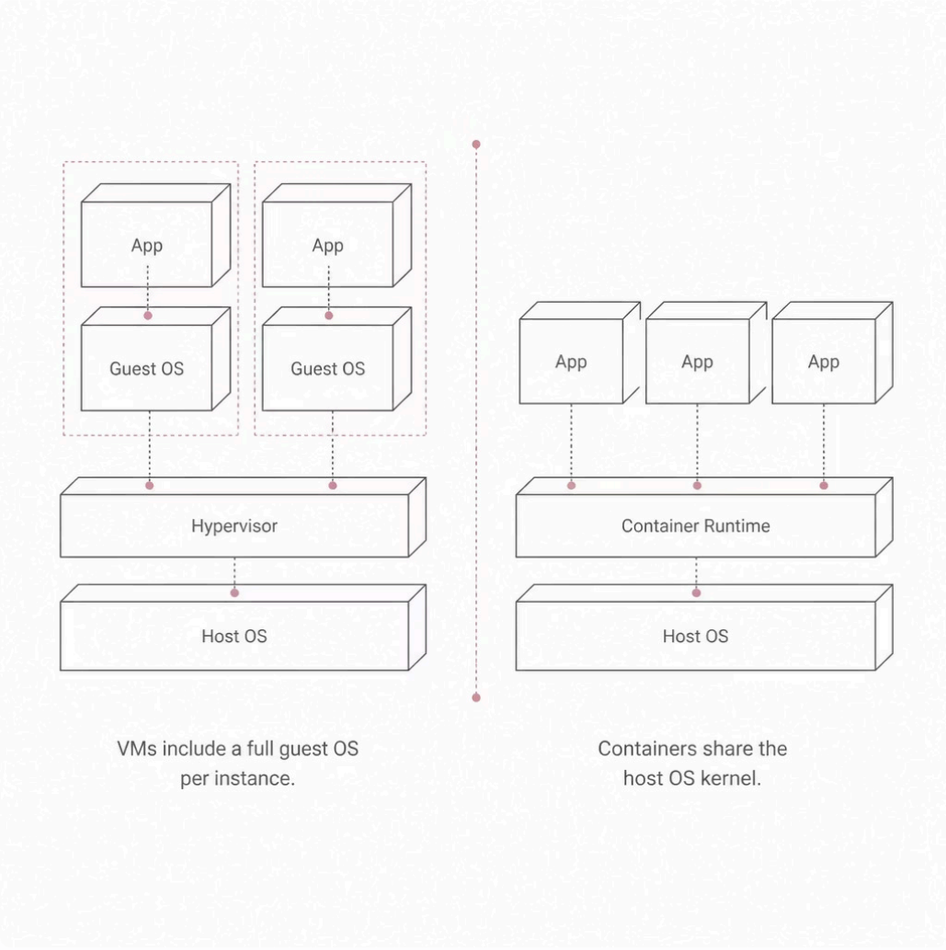
Virtual Machine vs. Container



Why Containers Win for Backend Services

Containers share the host OS kernel, making them lightweight and ideal for microservices.

Smaller

MBs vs GBs — no guest OS overhead

Faster

Seconds to start, not minutes

Portable

Same image runs on any host with Docker

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What Is Docker?

Docker is the most widely adopted container platform — the standard for packaging and shipping applications.

1. **Code**
   Write your application
2. **Dockerfile**
   Define build instructions
3. **Image**
   Immutable, shareable artifact
4. **Container**
   Running instance

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Docker Image vs. Container

🗂️ Image — The Blueprint

- Read-only template
- Built from a Dockerfile
- Reusable across environments
- Stored in a registry (e.g., Docker Hub)

Think of it as a **class definition**.

▶️ Container — The Running Instance

- Live, executing process
- Has state (memory, filesystem writes)
- Can start, stop, pause, restart
- Isolated from host and other containers

Think of it as an **object instantiated** from that class.

ⓘ One image → many containers

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Building Your First Dockerfile

```
FROM golang:1.24
WORKDIR /app
COPY . .
RUN go build -o mini-asm .
CMD ["./mini-asm"]
```

Line by Line

01

FROM

Base image — official Go 1.24

02

WORKDIR

Sets working directory inside container

03

COPY + RUN

Copy source, compile binary

04

CMD

Process that runs when container starts

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Docker Build & Run

1	2	3
Build the Image	Run the Container	Verify It Works
`docker build -t mini-asm .`	`docker run -p 8080:8080 mini-asm`	`curl http://localhost:8080/health`
Reads Dockerfile, produces a tagged image.	Starts a container, maps port 8080 to host.	Should return `200 OK` — same as local.

Containerized apps should behave **exactly** like local ones.

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Why Docker Compose?

Real applications are never a single process. A typical backend stack includes:

Managing each container manually with `docker run` is error-prone. Docker Compose defines all services in one file and starts them together.

```
services:
  api:
    build: .
    ports:
      - "8080:8080"
  postgres:
    image: postgres:17
    environment:
      POSTGRES_PASSWORD: secret
```

One command: `docker compose up`

API	Database	Cache	Message Queue

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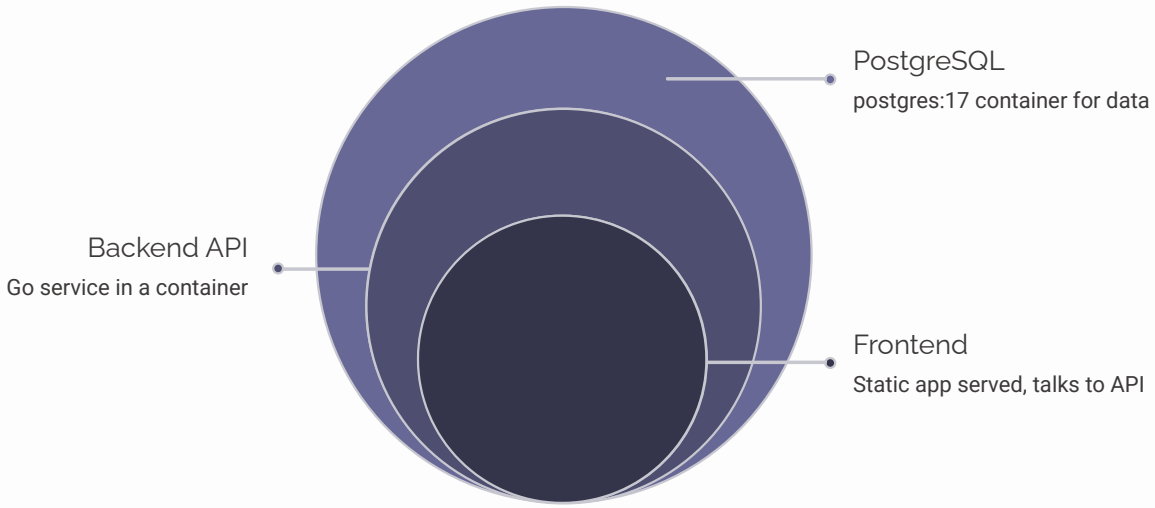
Containerizing Mini ASM



Mini ASM's architecture: frontend talks to a Go API container, which connects to a PostgreSQL container — all started with one command.

Today

`api` + `postgres` services in `docker-compose.yml`

Next

Add `redis` for caching, `scanner-worker` for async tasks

Key Takeaway

Containers standardize development and deployment — eliminating environment drift.